

# Skills Summary

## Vertices, Foci and Asymptotes of Ellipses and Hyperbolas

Use this reference while completing your worksheet or when studying.

Both curves are written with  $x^2$  and  $y^2$  over denominators. Which curve you have is decided by the sign between the two terms: a plus sign is an ellipse (a closed loop), a minus sign is a hyperbola (two branches). Which direction it runs is decided by the term that comes first, or is positive. The number under the transverse or major term is always  $a^2$ .

### Ellipse: vertices and foci from the equation

**Rule:** For  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  with  $a > b$ , the major axis is horizontal: vertices  $(\pm a, 0)$ , co-vertices  $(0, \pm b)$ , foci  $(\pm c, 0)$  with  $c^2 = a^2 - b^2$ .

**Subtract** for an ellipse: the foci sit *inside* the loop, so  $c < a$ . If the bigger denominator is under  $y^2$  instead, everything turns a quarter-turn: vertices  $(0, \pm a)$  and foci  $(0, \pm c)$ .

**Example:** Find the foci of  $\frac{x^2}{100} + \frac{y^2}{36} = 1$ .

**Step 1.** The bigger denominator, 100, is under  $x^2$ , so this is a horizontal ellipse and the foci are on the  $x$ -axis — use  $c^2 = a^2 - b^2$ .

**Step 2.**  $c^2 = 100 - 36 = 64$ , so  $c = \sqrt{64}$ .

Foci at  $(\pm c, 0)$  with  $c = 8$

**Try it:** Find  $c$  for  $\frac{x^2}{225} + \frac{y^2}{144} = 1$ .

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### Hyperbola: vertices and foci

**Rule:** For  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  the branches open left and right: vertices  $(\pm a, 0)$ , foci  $(\pm c, 0)$  with  $c^2 = a^2 + b^2$ .

**Add** for a hyperbola: the foci sit *beyond* the vertices, so  $c > a$  always. The positive term names the direction — if the  $y^2$  term is the positive one, the branches open up and down and  $a^2$  is the number under  $y^2$ .

**Example:** Find the vertices and foci of  $\frac{x^2}{9} - \frac{y^2}{16} = 1$ .

**Step 1.** The  $x^2$  term is positive, so the branches open sideways and both vertices and foci are on the  $x$ -axis — use  $c^2 = a^2 + b^2$ .

**Step 2.**  $a = \sqrt{9} = 3$ , so the vertices are  $(\pm 3, 0)$ ; and  $c^2 = 9 + 16 = 25$ .

Foci at  $(\pm c, 0)$  with  $c = 5$

**Try it:** Find  $c$  for  $\frac{x^2}{64} - \frac{y^2}{36} = 1$ .

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## Hyperbola: slopes of the asymptotes

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**Rule:** Draw the central rectangle:  $a$  each way along the transverse axis,  $b$  each way along the other. The asymptotes are its two diagonals extended.

Opening sideways:  $y = \pm \frac{b}{a}x$ . Opening up and down:  $y = \pm \frac{a}{b}x$ . Either way the number from under the *positive* term goes on the bottom when the curve opens sideways and on the top when it opens up and down. An ellipse has no asymptotes at all.

**Example:** Find the asymptote slopes of  $\frac{x^2}{25} - \frac{y^2}{4} = 1$ .

**Step 1.** The  $x^2$  term is positive, so this hyperbola opens sideways and the slopes are  $\pm \frac{b}{a}$  — rise  $b$  over run  $a$  on the central rectangle.

**Step 2.**  $a = \sqrt{25} = 5$  and  $b = \sqrt{4} = 2$ , so the slope is  $\frac{2}{5}$ , and the asymptotes are  $y = \pm \frac{2}{5}x$ .  
Slope =  $\frac{2}{5}$

**Try it:** Find the asymptote slope of  $\frac{x^2}{49} - \frac{y^2}{9} = 1$ .

$\frac{2}{5}$  :xpcəp

**(!) Watch out:** Do not read  $a$  as “the number under  $x^2$ ”. For an ellipse  $a^2$  is the *larger* denominator; for a hyperbola  $a^2$  is the denominator under the *positive* term. Getting that backwards inverts every asymptote slope. And never take the square root of the whole equation: the denominators are  $a^2$  and  $b^2$ , so take roots of those numbers alone.